

ATPL EASA STUDY GUIDE | 040 - HUMAN PERFORMANCE

Chapter 2: Aviation Physiology & Gas Laws

Syllabus Source: CAE Oxford (pp. 23-56) | Pass Target: >= 90% AviationExam

1. ATMOSPHERIC COMPOSITION & VERTICAL STRUCTURE

Clean, dry atmospheric air has an essentially constant volumetric composition up to approximately 70,000 ft (the homosphere):

- Nitrogen (N₂): 78.08% (~78%) - Inert diluent gas; dissolves in body fluids under pressure.
- Oxygen (O₂): 20.95% (~21%) - Vital gas for cellular metabolism and aerobic respiration.
- Argon (Ar): 0.93% - Inert noble gas.
- Carbon Dioxide (CO₂): 0.04% - Crucial physiological regulator of respiration.

ISA Standards at Mean Sea Level (MSL): Pressure = 1013.25 hPa (760 mmHg = 29.92 inHg); Temperature = +15°C. Standard temperature lapse rate: 2°C per 1,000 ft (6.5°C per 1,000 m) up to the Tropopause (36,090 ft / 11 km), where temperature becomes constant at -56.5°C.

2. THE MASTER AVIATION GAS LAWS (AVIATIONEXAM CORE)

Gas Law	Mathematical Formulation	Aeronautical & Human Physiological Impact
Dalton's Law	$P_{\text{total}} = P_1 + P_2 + \dots + P_n$	Total pressure equals the sum of partial pressures. Because O ₂ is 21%, $P_{\text{O}_2} = 0.21 \times P_{\text{ambient}}$. As ambient pressure falls with altitude, P_{O_2} falls proportionally, causing Hypoxic Hypoxia.
Boyle's Law	$P \times V = \text{Constant (at const T)}$	Pressure and volume of a gas are inversely proportional. As altitude increases, trapped body gases EXPAND (Middle ear, sinuses, tooth pulp, GI tract). Gas volume doubles at FL 180 (0.5 atm)!
Henry's Law	Amount dissolved = $k \times P_{\text{gas}}$	The mass of gas dissolved in a liquid is proportional to its partial pressure. Rapid decompression causes nitrogen to bubble out of physical solution, causing Decompression Sickness (DCS).
Graham's Law	Rate of diffusion inversely prop. to $\sqrt{\text{mass}}$	Gases diffuse from high partial pressure to low partial pressure across alveolar-capillary membranes. Light gases diffuse faster.
Charles' Law	$V_1 / T_1 = V_2 / T_2$ (at const P)	Gas volume increases directly with temperature. Critical for oxygen storage cylinder pressure variations with ambient temperature.

[!] EXAM TRAP (AVIATIONEXAM) - Gas Volume Expansion Factors with Altitude (Boyle's Law)

AviationExam frequently tests the exact volume multiplier for trapped body gases: -
 Sea Level (1.0 atm / 1013 hPa): Volume = 1.0 - FL 180 (18,000 ft / 0.5 atm / ~500 hPa): Gas volume DOUBLES (2.0x) - FL 340 (34,000 ft / 0.25 atm / ~250 hPa): Gas volume QUADRUPLES (4.0x) - FL 420 (42,000 ft / 0.17 atm / ~170 hPa): Gas volume expands SIX TIMES (6.0x)!

3. RESPIRATORY MECHANICS & ALVEOLAR GAS EXCHANGE

- External Respiration: Gas exchange in the lungs between alveoli and pulmonary capillary blood. - Internal Respiration: Gas exchange between systemic capillaries and metabolizing tissue cells. - Alveolar Gas Environment: In the alveoli, inspired air is fully humidified with water vapor ($P_{H_2O} = 47 \text{ mmHg}$ at 37°C) and mixed with carbon dioxide ($P_{CO_2} = 40 \text{ mmHg}$).

[*] OFFICIAL DEFINITION - The Alveolar Oxygen Formula (Critical Physiological Threshold)

$P_{A_{O_2}} = (P_{\text{ambient}} - 47) \times 0.21 - (P_{CO_2} / 0.8)$ - At Sea Level: $P_{A_{O_2}} = (760 - 47) \times 0.21 - (40 / 0.8) = 150 - 50 = \sim 100 \text{ mmHg}$. - At 10,000 ft: $P_{A_{O_2}}$ drops to $\sim 60 \text{ mmHg}$ (arterial saturation drops to 90%). Performance deteriorates! - Above 63,000 ft (Armstrong's Line): Ambient pressure drops below 47 mmHg; body fluids boil at 37°C !

4. HEMOGLOBIN, OXYGEN TRANSPORT & THE BOHR EFFECT

Oxygen is carried in the blood in two forms: 1.5% physically dissolved in blood plasma, and 98.5% chemically bound to hemoglobin (Hb) inside erythrocytes (red blood cells). Each gram of Hb can bind 1.34 ml of oxygen.

- Oxygen Dissociation Curve: A non-linear, sigmoidal (S-shaped) curve relating arterial P_{O_2} to hemoglobin saturation (SaO_2). The steep portion below 60 mmHg means small altitude increases cause rapid desaturation.
- The Bohr Effect (Right Shift): The curve shifts to the RIGHT (lowering Hb affinity for oxygen, promoting O_2 unloading to tissues) under: Increased blood P_{CO_2} (hypercapnia), Increased acidity (lower pH), Increased body temperature, and increased 2,3-DPG.

5. THE PRIMARY CHEMICAL DRIVER OF RESPIRATION

[!] EXAM TRAP (AVIATIONEXAM) - Absolute Exam Trap: What Controls Your Breathing Rate?

- Primary Chemical Stimulus: Breathing rate and depth are controlled by chemoreceptors in the MEDULLA OBLONGATA sensitive to the partial pressure of CARBON DIOXIDE (P_{CO_2}) and H^+ ion concentration (pH) in arterial blood. - Secondary Stimulus: Peripheral chemoreceptors in the carotid and aortic bodies monitor low P_{O_2} (hypoxia), but this hypoxic drive acts only as a backup when P_{O_2} drops below 60 mmHg!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): SHELL Model & Error Analysis

REGLA MNEMOTÉCNICA: L-H-S-E-L Paso 1: Identificar siempre el Liveware Central (L) como el piloto/operador humano. Paso 2: Si el problema involucra mandos, pantallas o botones confusos -> Es un desajuste L-H (Hardware). Paso 3: Si involucra una checklist mal redactada o un manual confuso -> Es un desajuste L-S (Software). Paso 4: Si involucra turbulencia, calor o hipoxia en cabina -> Es un desajuste L-E (Environment). Paso 5: Si involucra mala coordinación entre Capitán y Primer Oficial o con ATC -> Es un desajuste L-L (Liveware).

6. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the primary chemical stimulus that regulates the rate and depth of breathing in a healthy human?

- [A]: Low oxygen partial pressure in the lungs

- [B]: High nitrogen saturation in the blood
- [C]: The concentration of carbon dioxide (P_{CO_2}) and hydrogen ions in arterial blood
- [D]: Blood glucose concentration

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. The respiratory center in the brainstem (medulla) is primarily driven by P_{CO_2} and pH levels, NOT by oxygen deficiency.

Q2: According to Boyle's law, if an aircraft experiences a rapid decompression from sea level to FL 180, what happens to the volume of gas in the lungs?

- [A]: It remains unchanged
- [B]: It doubles (2x)
- [C]: It quadruples (4x)
- [D]: It is halved (0.5x)

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. At FL 180, atmospheric pressure is halved (~500 hPa vs 1013 hPa), so gas volume doubles according to Boyle's law ($P \times V = \text{const}$).

Q3: Which gas law explains the development of Decompression Sickness (DCS) during high altitude flight?

- [A]: Dalton's Law
- [B]: Boyle's Law
- [C]: Henry's Law
- [D]: Charles' Law

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. Henry's law states that the amount of dissolved gas in a liquid depends on its partial pressure. When pressure drops, dissolved nitrogen bubbles out.

Q4: What is the percentage of oxygen in atmospheric air at 34,000 ft compared to sea level?

- [A]: Exactly the same (approx. 21%)
- [B]: Approximately 10%
- [C]: Less than 5%
- [D]: 0%

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [A]. The percentage of oxygen remains constant at ~21% throughout the homosphere (up to 70,000 ft). Hypoxia occurs because total pressure, and therefore oxygen partial pressure, drops (Dalton's Law).

Q5: What is 'Armstrong's Line' in aviation physiology and at what approximate altitude does it occur?

- [A]: FL 100 where supplemental oxygen is required
- [B]: FL 180 where gas volume doubles
- [C]: Approx. 63,000 ft where atmospheric pressure equals body water vapor pressure (47 mmHg) and exposed body fluids boil
- [D]: The tropopause at 36,000 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. At 63,000 ft (pressure 47 mmHg / 63 hPa), water boils at normal body temperature (37°C). A full pressure suit is mandatory.

Q6: Which condition shifts the hemoglobin oxygen dissociation curve to the right (Bohr Effect), facilitating oxygen delivery to tissues?

- [A]: Decreased temperature and hypothermia
- [B]: Decreased P_{CO2} (hypocapnia) and elevated pH (alkalosis)
- [C]: Increased blood P_{CO2}, increased acidity (lower pH), and elevated temperature
- [D]: Low altitude exposure

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. Acidosis, high P_{CO2}, and heat shift the curve rightwards, decreasing Hb affinity and facilitating oxygen delivery to metabolizing tissues.

Q7: What is the standard atmospheric pressure and temperature at Mean Sea Level according to the International Standard Atmosphere (ISA)?

- [A]: 1000 hPa and 20°C
- [B]: 1013.25 hPa (760 mmHg) and +15°C
- [C]: 1020 hPa and 0°C
- [D]: 29.92 hPa and 15°C

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Standard ISA MSL values are 1013.25 hPa (29.92 inHg / 760 mmHg) and +15°C.

Q8: Under what physiological condition does the body's peripheral chemoreceptor backup drive (hypoxic drive) become a significant factor in breathing?

- [A]: When arterial P_{O2} falls below approximately 60 mmHg (corresponding to altitudes above 10,000 ft)
- [B]: When blood pH rises above 7.45
- [C]: Only during deep REM sleep
- [D]: When nitrogen bubbles form in capillaries

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [A]. The peripheral chemoreceptors (carotid and aortic bodies) respond to severe hypoxia only when arterial P_{O2} falls below ~60 mmHg.